

Progress Report Sprint 3

- Course Name: CS461 Senior Software Engineering Project I
- Date: 11/05/25
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
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Summary

- Continued work on web application/UI layout and design
- Implemented javascript movement and input handling
- Created a microservice to interact with user input

Progress vs Plan

PLANNED ITEM	STATUS (% OR DONE/PARTIAL/NOT DONE)	LINKS (ISSUE/PR)	NOTES/BLOCKERS	PLAN CHANGE
WEB DESIGN	Partial	https://github.com/marknovak/TaxonBodyMassML/pull/2	I want to do some research into color schemes and font styles before doing more css styling.	N/A
PREDICTIVE ALG	Not Done		We decided it's too early to spend too much time on this right now.	Focus on this after data split optimization, normalization, and merge.
TRAINING/TESTING DATA SPLIT	Partial	https://github.com/marknovak/TaxonBodyMassML/pull/1	Changed approach to use random sampling and z-normalization instead of the previous multi-leveled approach.	N/A
SPECIES NAME LOOKUP	Partial	https://github.com/marknovak/TaxonBodyMassML/tree/web_dev	Created python microservice which serves as a prototype for the species name lookup. Returns mass_g if name exactly matches.	This is a temporary solution and will be replaced later. Will later return full taxonomy as well.

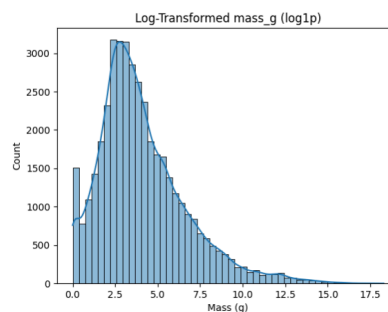
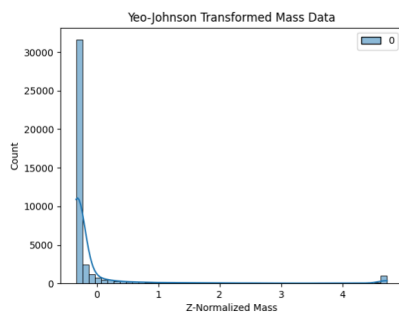
CONTINUOUS INTEGRATION	Partial	https://github.com/marknovak/TaxonBodyMassML/blob/main/.github/workflows/CI.yml	Implemented linting on the main branch.	N/A
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Evidence of Working Software

Web Development latest update commit hash: 6bf081b9c567bf952b63515fe30602a4d88b23d7

Link to UI: https://praterh.github.io/HaileysTaxonBodyMassML/web_dev/index.html

Home	Data Visualization	Citations	Help
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Risk and Quality

- We are required to show an end-to-end slice of our program, but we do not have our algorithm yet. In order for us to be able to test the functionality of our web application, we need some temporary function to serve as a placeholder for our predictive algorithm.
 - Owner: Grant
 - Status: Complete
 - Concrete actions: The best course of action for us right now was to implement a simple microservice. The microservice is the `prototype_lookup.py` file, which is a flask app. It processes the `BodyMass.csv` data into a data frame and receives GET requests via the `/single_species` endpoint, returning a mass if it exists in the database.
- The web-dev branch had pulled from the data-partitioning branch earlier on in the term to access the histograms that would be displayed on the UI. Because of this, there will be a lot of conflicts when merging with the main branch (which had since been merged with the updated data-partitioning branch).
 - Owner: Hailey
 - Status: Complete
 - Concrete Actions: I worked closely with Grant on this because he was the primary developer of the data-partitioning branch. We discussed where we would keep updates from the web-dev branch and where it would make the most sense to remove files altogether. For instance, there were copies of `test.csv` and `train.csv` in the web-dev branch that were severely out-of-date; these files are very large, so we decided it would be the best course of action to simply delete them from the web-dev branch before merging to main.
- When merging the body mass data set and the catalog of life data, the outputted data file might not match the taxonomy and the species mass accurately. The risk here is that a incorrect mass might be matched to the incorrect species ruining the data.
 - Owner: Anish
 - Status: In-progress

- Concrete Actions: To ensure the correct matching of species with their body mass data, preprocess the species names in both datasets to remove extra spaces and standardize them to lowercase. Use fuzzy matching with the fuzzywuzzy library to match species names that may differ slightly between datasets and only accept matches with a high confidence score (e.g., 90%). Finally, merge the datasets based on the best-matched species names and save the resulting data, ensuring the mass is correctly assigned to each species.

Next Goals

- Further the css styling of the UI; this will work toward REQ-006, meeting usability and accessibility standards
- Begin to work on writing our predictive algorithm; this will allow for progress toward REQ-001 & REQ-002
- Finalize the merging and cleaning of our datasets; this has a connection to REQ-007 because both of these sets are publicly accessible

Individual Contributions

Grant:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/data-partitioning>
- Implemented z-normalization and improved visualizations.
- Created a prototype lookup for masses as a microservice for the web dev side of the project.
- Configured super-linter for contiguous integration using GitHub actions.
- Recorded the walking prototype demo video.

Hailey:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/web-dev>
- Talked to and worked with project partner on more in depth layout of website
- Cleaned up and finalized the main page template
- Incorporated javascript clicking behavior into UI
- Introduced framework for future microservice implementation into front-end javascript
- Deployed Grant's microservice using render for use on web application

Anish:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/data-combination> and <https://github.com/marknovak/TaxonBodyMassML/tree/test-cases>
- Worked on making test cases for walking skeleton prototype
- Setup testing environment
- Cleaned up some data from the Open Tree of Life source data

Team Reflection

This week, we did some good work on our user interface and initialized the code that will later contain our machine-learning algorithm. We are happy with what we got done this week, but we know that there are some things that we can improve upon as the year continues.

We should have been merging to the main branch much earlier than this week. Furthermore, we assumed that it would be easier to remain working in our own branches until merging to main was necessary, but that proved to make things much more difficult for us this week. Similarly, we will definitely be much more aware about our formatting and commenting practices; we had to spend a good amount of time on those simple things this week when they should have just been getting done over time.

Now, we're excited to begin working on our predictive algorithm with these criticisms in mind and, in turn, produce better work overall.